

We selected ten small and medium enterprises (SMEs) in and around Bengaluru and Hosur to participate in this project. People from these enterprises were completely trained by our experts and qualified as an internal auditor. With our training, they built the in-house capacity. We also helped them identify the gap in their existing system.

What's the selection criterion for SMEs to participate in this project?

Basically, the organisation should deal with electrical or electronics equipment. Or, it can be a supplier to these industries. Also, it must be ISO 9000 certified.

Is HSPM compulsory in India?

HSPM is not compulsory even across the globe, but it will help you meet some requirements of the regulation in Europe.

So if you are not HSPM certified, you will not be able to export to Europe, right?

That's correct. In China, 1500 companies have already been certified. It is very active in Taiwan, the whole of Europe and the US because all these countries already have their restriction of hazardous substances related regulations in place.

What should ISO 9001 certified companies do?

At the moment, most companies are ISO 9000 certified, 14000 certified or even 27000 certified, but this particular standard will help them control the use of hazardous substances. Their quality policy, management objectives, and communications and management representatives will have a specific method to control the hazardous substances. They even have to plan the hazardous substance assessment and control at a regular interval. So the entire activity of the organisation will ensure that the HSPM is implemented thoroughly. That is the reason why this standard works on top of ISO 9001.

Most of the ISO 9001 clauses are common in QCO 80000. But it is not practically possible to analyse every product or sample. It is time-consuming as well as capital-intensive. Rather, a system-level approach where the system doesn't allow processing or procurement of any of the hazardous substance materials is ideal. This standard will help compa-

nies to migrate from the existing level to that of a particular hazardous substance-free level.

Which is the most extensively used hazardous substance?

In electronics industry, lead is extensively used in the form of leaded solders. In switching over from leaded solder to lead-free solder, manufacturers face a unique problem called 'tin whiskering.' The tin-base or silver-base solder used in place of leaded solder causes a whisker growth or needle-type growth. This needle growth leads to some premature failures.

We help manufacturers who have switched to lead-free solder technology, by providing them solutions to reliability and performance issues with the help of various simulation chambers and scanning electron microscope. Of course, it is not part of the project but we fully build the in-house capacity and expertise to handle the switch-over. This includes testing their components and products for reliability under different climatic conditions, safety level test, material level test and ensuring that the product will perform as good as it was performing with leaded solder.

How do you convince companies to move from leaded solder to lead-free solder technology?

When moving from leaded solders to lead-free solders, there is a 20°C rise in melting point. So the complete soldering operation will occur at a temperature 20°C higher than the normal operating point. Secondly, other parts need not be changed. Only the compatible surface will be exposed to 20°C higher temperature.

In addition to that, in certain extreme environments these lead-free solders have a tendency to form tin whisker. When tin whisker grows, it leads to some sort of short-circuit on the boards. So if a safety-critical device like automotive electronic component or medical electronics device is designed using a lead-free solder, this unwanted growth could lead to malfunctioning due to short-circuiting and a lot of premature failures.

So it is very important to tackle these things during the design itself. That's where TÜV Rheinland plays a vital role to help these companies switch over from leaded technology to lead-free technology. ●

(The complete interview can be viewed at electronicsforu.com)

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